THE SOLUTIONS

1. (a) A line L through the origin has direction vector $\underline{\mathbf{b}} = (1,1,1)^T$. Find the projection of the vector $\underline{\mathbf{a}} = (1,2,3)^T$ onto/along the vector $\underline{\mathbf{b}}$. Use this to find a point Q on L, nearest to the point $P(1,2,3)$ and hence the distance from P to L. Provide a sketch and explain your reasoning! [5]

Solution: Projection is given by $\lambda\underline{\mathbf{b}}$ where

$$\lambda = \frac{\underline{\mathbf{a}}\cdot\underline{\mathbf{b}}}{\underline{\mathbf{b}}\cdot\underline{\mathbf{b}}} = \frac{1+2+3}{1+1+1} = 2$$

so the projection is the vector $(2,2,2)^T$

As $\underline{\mathbf{a}} = \overrightarrow{OP}$, we see (sketch at end) that the origin, P and Q form a right angled triangle, with sides $\underline{\mathbf{a}}$ and $\lambda\underline{\mathbf{b}}$, so the point on L nearest to P is given by the projection, with coordinates $Q(2,2,2)$. Hence, the vector $\overrightarrow{PQ}$ is $(1,0,-1)^T$ and the distance is $|\overrightarrow{PQ}| = \sqrt{2}$.

(b) Show that both $u(x,y) = \sin(x^2-y^2)$ and $v(x,y) = \cos(y^2-x^2)$ are possible solutions for w in the partial differential equation (PDE)

$$y\frac{\partial w}{\partial x} + x\frac{\partial w}{\partial y} \;=\; 0\,.$$

Hence show that the function $z(x,y) \;=\; x^2u - y^2v$ satisfies the PDE

$$y\frac{\partial z}{\partial x} + x\frac{\partial z}{\partial y} \;=\; 2xy(u-v)\,.$$

[7]

Solution: (i) The derivatives are

$$u_x = 2x\cos(x^2-y^2)\,,\; u_y = -2y\cos(x^2-y^2)\,,\; v_x = 2x\sin(y^2-x^2)\,,\quad \text{and}\quad v_y = -2y\sin(y^2-x^2)\,.$$

Hence

$$xu_y + yu_x \;=\; x\left[-2y\cos(x^2-y^2)\right] + y\left[2x\cos(x^2-y^2)\right] = 0\,,$$

and

$$xv_y + yv_x \;=\; x\left[-2y\sin(y^2-x^2)\right] + y\left[2x\sin(y^2-x^2)\right] = 0\,,$$

so both satisfy the equation $yw_x + xw_y = 0$, as required.

(ii) Given $z \;=\; x^2u - y^2v$ we have

$$z_x = 2xu + x^2u_x - y^2v_x\,,\quad \text{and}\quad z_y = x^2u_y - 2yv - y^2v_y\,,$$

so that

$$yz_x + xz_y = y(2xu + x^2u_x - y^2v_x) + x(x^2u_y - 2yv - y^2v_y) = 2xy(u-v) + x^2(yu_x + xu_y) - y^2(yv_x + xv_y) = 2xy(u-v)\,,$$

using (i).

(c) A function of two variables is given as

$$f(x,y) \;=\; x^2 \cos y \,.$$

(i) Obtain the Taylor series expansion of $f(x,y)$ about the point $(x_0,y_0) = (1,\pi)$, up to quadratic terms.

(ii) If we use this expansion to estimate $(1.1)^2\cos(3)$, consider the discarded terms of order three and higher, to give an upper bound for the size of the expected error. Explain your reasoning! [6]

Solution: (i) $f(1,\pi) = -1$. We need the derivatives

$$f_x = 2x\cos y \;,\; f_y = -x^2 \sin y \;,\; f_{xx} = 2\cos y \;,\; f_{yy} = -x^2\cos y \;, \quad \text{and} \quad f_{xy} = -2x\sin y$$

Hence, evaluated at $(1,\pi)$ we get the values

$$f_x = -2 \;,\; f_y = 0 \;,\; f_{xx} = -2 \;,\; f_{yy} = 1 \;, \quad \text{and} \quad f_{xy} = 0$$

and letting $h = x - x_0$ and $k = y - y_0$, the Taylor series to second order is given as

$$f(x,y) = f(x_0,y_0) + hf_x(x_0,y_0) + kf_y(x_0,y_0) + \frac{1}{2}\left[h^2 f_{xx}(x_0,y_0) + 2hkf_{xy}(x_0,y_0) + k^2 f_{yy}(x_0,y_0)\right] + \cdots$$

hence we have

$$x^2\cos y \;=\; -1 - 2(x-1) + \frac{1}{2}[-2(x-1)^2 + (y-\pi)^2] + \cdots \;=\; -1 - 2(x-1) - (x-1)^2 + \frac{1}{2}(y-\pi)^2 + \cdots$$

(ii) The size (absolute value) of the discarded terms is given by powers of h and k up to order three: h^3, hk^2, h^2k and k^3. For $x = 1.1, y = 3$, we have $h = 0.1$ and $k = 3 - \pi \approx -0.1$. Thus all four terms of order three will be of the form $\pm(0.1)^3$, and the best bound for the size of the error is 10^{-3}.

(d) Show that the first-order Ordinary Differential Equation

$$\frac{2x}{y} - \cos x \;=\; \left(\frac{x^2}{y^2} - 2e^{2y}\right)\frac{dy}{dx}$$

is exact and find a solution satisfying $y(0) = -1$. [7]

Solution: Rearrange the ODE to

$$\left(\frac{x^2}{y^2} - 2e^{2y}\right)\frac{dy}{dx} + \cos x - \frac{2x}{y} \;=\; 0$$

and identify $P(x,y) = \dfrac{x^2}{y^2} - 2e^{2y}$ and $Q(x,y) = \cos x - \dfrac{2x}{y}$, as in formula sheet. For an exact equation we require

$$\frac{\partial P}{\partial x} \;=\; \frac{\partial Q}{\partial y} \,.$$

Both derivatives are equal to $\dfrac{2x}{y^2}$, so the ODE is exact and a function $f(x,y)$ exists with

$$\frac{\partial f}{\partial x} \;=\; Q(x,y) \;=\; \cos x - \frac{2x}{y} \Rightarrow f(x,y) = \int \cos x - \frac{2x}{y}\,dx \;=\; \sin x - \frac{x^2}{y} + A(y) \,,$$

and

$$\frac{\partial f}{\partial y} \;=\; P(x,y) \;=\; \frac{x^2}{y^2} - 2e^{2y} \Rightarrow f(x,y) = \int \frac{x^2}{y^2} - 2e^{2y}\, dy \;=\; -\frac{x^2}{y} - e^{2y} + B(x)$$

where A and B are arbitrary functions. Comparing the two, we have $A(y) = -e^{2y}$ and $B(x) = \sin x$, so the solution of the ODE is given by $f =$constant:

$$-\frac{x^2}{y} - e^{2y} + \sin x = C$$

and with the initial condition $y(0) = -1 \Rightarrow C = -e^{-2}$, so

$$\frac{x^2}{y} + e^{2y} - \sin x = \frac{1}{e^2}$$

2. (a) Evaluate the double integral

$$I = \int\int_R xy\, e^{x^2+y^2}\, dx\, dy\,,$$

(i) when R is a square with vertices at the origin, $(1,0),(0,1)$, and $(1,1)$; [4]

(ii) when R is in the first quadrant, bounded by the coordinate axes, and the unit circle $x^2+y^2=1$, changing to polar coordinates (r,θ) . You may assume the result $dxdy = rdrd\theta$ [5]

Solution: (i) The integral over the square can be written as

$$I = \int_0^1\int_0^1 xy\, e^{x^2+y^2}\, dx\, dy = \int_0^1\int_0^1 xy\, e^{x^2}e^{y^2}\, dx\, dy = \left(\int_0^1 x\, e^{x^2}\, dx\right)\left(\int_0^1 y\, e^{y^2}\, dy\right)$$

so we only need to integrate one of them, as they are identical:

$$\int_0^1 x\, e^{x^2}\, dx = \left[\frac{1}{2}e^{x^2}\right]_0^1 = \frac{1}{2}(e-1) \Rightarrow I = \frac{1}{4}(e-1)^2\,.$$

(ii) In polar coordinates $x = r\cos\theta$ and $y = r\sin\theta$ the integrand becomes

$$xye^{x^2+y^2} = (r\cos\theta)(r\sin\theta)e^{r^2} = r^2\frac{1}{2}\sin(2\theta)e^{r^2}\,.$$

Over the quarter-circle, $r = 0\ldots1$ and $\theta = 0\ldots\dfrac{\pi}{2}$, and using the Jacobian given, we can rewrite the integral as

$$I = \frac{1}{2}\int_{\theta=0}^{\pi/2}\int_{r=0}^1 \sin(2\theta)\left(r^2e^{r^2}\right) r\, dr\, d\theta = \frac{1}{2}\int_{\theta=0}^{\pi/2}\sin(2\theta)\, d\theta\int_{r=0}^1 r^3e^{r^2}\, dr$$

The first integral is straightforward:

$$\int_{\theta=0}^{\pi/2}\sin(2\theta)\, d\theta = -\frac{1}{2}\left[\cos 2\theta\right]_0^{\pi/2} = 1$$

and integrating by parts,

$$\int_{r=0}^1 r^3e^{r^2}\, dr = \left[\frac{1}{2}r^2e^{r^2} - \int re^{r^2}\, dr\right]_0^1 = \left[\frac{1}{2}r^2e^{r^2} - \frac{1}{2}e^{r^2}\right]_0^1 = \frac{1}{2}$$

so $I = \dfrac{1}{2}(1)\dfrac{1}{2} = \dfrac{1}{4}$.

(b) Let a function be defined as $f(x) = -x-1$ on the interval $[-1,0]$.

(i) Sketch $Eve(x)$ and $Odd(x)$, the even and odd periodic extensions of f, both with period $T=2$, for $-3 \le x \le 3$. [2]

(ii) If Fourier Series for $Eve(x)$ and $Odd(x)$ are calculated, what is the value each Fourier Series takes at $x = 2$? State the rate of decay of the Fourier coefficients for each series. [3]

(iii) Obtain the Fourier Series for $Eve(x)$. [6]

(iv) Use (iii) and Parseval's Theorem to show that $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n^4} = \frac{\pi^4}{90}$ [5]

Solution: (i) Sketch at end.

(ii) The odd extension has a discontinuity at $x = 2$ so the series converges to the average of the values at either end:

$$\frac{1}{2}\left(Odd(x_-) + Odd(x_+)\right) = \frac{1}{2}(-1+1) = 0\,.$$

The presence of the discontinuity in $Odd(x)$ gives that the Fourier coefficients decay as $1/n$. The even extension has no discontinuity so the series converges to the value of the function $Eve(2) = -1$, and Fourier coefficients decay as $1/n^2$.

(iii) The function is even, so $b_n = 0$. Now for $T = 2$,

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} Eve(x)\cos(2n\pi x/T)\,dx \;=\; \int_{-1}^{1} Eve(x)\cos(n\pi x)\,dx \;=\; 2\int_0^1 (x-1)\cos(n\pi x)\,dx$$

using half-range approach. Hence, integrating by parts, we have

$$a_n = 2\left[(x-1)\frac{\sin(n\pi x)}{n\pi}\right]_0^1 - \frac{2}{n\pi}\int_0^1 \sin(n\pi x)\,dx = 0-0-\frac{2}{n\pi}\left[-\frac{\cos(n\pi x}{n\pi}\right]_0^1 = \frac{2}{n^2\pi^2}[\cos(n\pi)-1] = -\frac{4}{n^2\pi^2}$$

when n is odd, and zero when n is even. The constant is calculated separately

$$a_0 = 2\int_0^1 (x-1)\,dx \;=\; -1\,.$$

The Fourier Series is

$$Eve(x) \;=\; -\frac{1}{2} - \frac{4}{\pi^2}\sum_{\substack{n=1\\ n\ odd}}^{\infty}\frac{1}{n^2}\cos(n\pi x)\,.$$

(iv) With $T = 2$, Parseval's theorem has

$$\int_{-1}^{1} f^2(x)\,dx \;=\; \frac{1}{2}a_0^2 + \sum_{n=1}^{\infty} a_n^2 \;=\; \frac{1}{2}(-1)^2 + \sum_{\substack{n=1\\ n\ odd}}^{\infty}\left(-\frac{4}{n^2\pi^2}\right)^2 \;=\; \frac{1}{2} + \frac{16}{\pi^4}\sum_{\substack{n=1\\ n\ odd}}^{\infty}\frac{1}{n^4}\,.$$

The left-hand side gives

$$\int_{-1}^{1} f^2(x)\,dx = 2\int_0^1 (x-1)^2\,dx \;=\; 2\left[\frac{1}{3}(x-1)^3\right]_0^1 \;=\; \frac{2}{3} \;\Rightarrow\; \sum_{\substack{n=1\\ n\ odd}}^{\infty}\frac{1}{n^4} \;=\; \frac{\pi^4}{16}\left(\frac{2}{3}-\frac{1}{2}\right) \;=\; \frac{\pi^4}{96}$$

Given that we require

$$S = \sum_{n=1}^{\infty}\frac{1}{n^4} \;=\; \sum_{\substack{n=1\\ n\ odd}}^{\infty}\frac{1}{n^4} + \sum_{\substack{n=1\\ n\ even}}^{\infty}\frac{1}{n^4}$$

we let $n = 2m$ in the second sum and get

$$S = \sum_{n=1}^{\infty}\frac{1}{n^4} \;=\; \frac{\pi^4}{96} + \sum_{m=1}^{\infty}\frac{1}{(2m)^4} \;=\; \frac{\pi^4}{96} + \frac{1}{16}\sum_{m=1}^{\infty}\frac{1}{(m)^4} \;=\; \frac{\pi^4}{96} + \frac{1}{16}S$$

so that $S(1-1/16) = \dfrac{\pi^4}{96} \Rightarrow S = \dfrac{16}{15}\cdot\dfrac{\pi^4}{96} \;=\; \dfrac{\pi^4}{90}$.

3. (a) Show (prove) that the Fourier transform of a cosine function is

$$\mathscr{F}[\cos(at)] = \pi[\delta(\omega-a)+\delta(\omega+a)]$$

where $\delta(x)$ is the Dirac delta-function. Derive a similar expression for the Fourier transform of $\sin(at)$.[7]

Solution: Apply the inverse transform in the first case:

$$\mathscr{F}^{-1}\left[\pi\Big(\ \delta(\omega-a)+\delta(\omega+a)\ \Big)\right] \ = \ \frac{1}{2\pi}\int_{-\infty}^{\infty}\pi\Big(\ \delta(\omega-a)+\delta(\omega+a)\ \Big)e^{i\omega t}\,d\omega$$

$$= \ \frac{1}{2}\int_{-\infty}^{\infty}\delta(\omega-a)e^{i\omega t}\,d\omega \ + \ \frac{1}{2}\int_{-\infty}^{\infty}\delta(\omega+a)e^{i\omega t}\,d\omega \ = \ \frac{1}{2}\left(e^{i(a)t}+e^{i(-a)t}\right),$$

where we have used the sampling property of the delta-function at $\omega = \pm a$ in the last step. Hence

$$\mathscr{F}^{-1}\left[\pi\Big(\ \delta(\omega-a)+\delta(\omega+a)\ \Big)\right] \ = \ \cos(at),$$

so that

$$\mathscr{F}[\cos(at)] = \pi[\delta(\omega-a)+\delta(\omega+a)]\ ,$$

as required.

To get a sine, we need $e^{iat}-e^{-iat}$ from the inverse transform, so we try

$$\mathscr{F}^{-1}\left[\pi\Big(\ \delta(\omega-a)-\delta(\omega+a)\ \Big)\right] = \frac{1}{2\pi}\int_{-\infty}^{\infty}\pi\Big(\ \delta(\omega-a)-\delta(\omega+a)\ \Big)e^{i\omega t}\,d\omega = \frac{1}{2}\left(e^{i(a)t}-e^{i(-a)t}\right) \ = \ i\sin(at)$$

so that

$$\mathscr{F}[\sin(at)] = -i\pi[\delta(\omega-a)-\delta(\omega+a)]\ .$$

(b) (i) Use the definition of the Laplace transform to prove the first shift theorem, that if $\mathscr{L}[f(t)] \ = \ \bar{f}(s)$, then

$$\mathscr{L}[e^{at}f(t)] \ = \ \bar{f}(s-a)\ . \qquad [\,4\,]$$

Solution:

$$\mathscr{L}[e^{at}f(t)] \ = \ \int_0^{\infty}e^{at}f(t)e^{-st}\,dt \ = \ \int_0^{\infty}f(t)e^{-(s-a)t}\,dt$$

Let the constant $s-a=\sigma$, then

$$\int_0^{\infty}f(t)e^{-\sigma t}\,dt \ = \ \bar{f}(\sigma) \ = \ \bar{f}(s-a)\ ,$$

as required.

(ii) Use partial fractions and the first shift theorem to obtain the following inverse Laplace transform:

$$\mathscr{L}^{-1}\left[\frac{10}{(s-2)(s^2+2s+2)}\right]\ . \qquad [\,7\,]$$

Solution: Partial fractions give

$$\frac{10}{(s-2)(s^2+2s+2)} \ = \ \frac{A}{s-2}+\frac{Bs+C}{s^2+2s+2} \ = \ \frac{A(s^2+2s+2)+(s-2)(Bs+C)}{(s-2)(s^2+2s+2)}$$

and equating numerators we have $10 = A(s^2+2s+2)+(s-2)(Bs+C)$.
Let $s = 2 \Rightarrow 10 = 10A \Rightarrow A = 1$.
Let $s = 0 \Rightarrow 10 = 2A - 2C \Rightarrow C = -4$.
Finally, any other value of s will do, to find $B = -1$. Hence

$$\mathscr{L}^{-1}\left[\frac{10}{(s-2)(s^2+2s+2)}\right] = \mathscr{L}^{-1}\left[\frac{1}{s-2} - \frac{s+4}{(s+1)^2+1}\right] = e^{2t} - \mathscr{L}^{-1}\left[\frac{s+1}{(s+1)^2+1} + \frac{3}{(s+1)^2+1}\right]$$

and the first shift theorem gives

$$= e^{2t} - e^{-t}\mathscr{L}^{-1}\left[\frac{s}{s^2+1} + \frac{3}{s^2+1}\right] = e^{2t} - e^{-t}(\cos t + 3\sin t) \ .$$

(iii) Use Laplace transforms and (ii), to solve the Ordinary Differential Equation

$$\frac{d^2x}{dt^2} - \frac{dx}{dt} - 2x = 10e^{-t}\cos t$$

which satisfies the initial conditions $x(0) = 1$ and $x'(0) = 3$. [7]

Solution: Take transforms using the formula sheet. For the RHS, use the first-shift theorem:

$$s^2\bar{x} - sx(0) - x'(0) - [s\bar{x} - x[0] - 2\bar{x} = \left[\frac{10s}{s^2+1}\right]_{s\to s+1} = \frac{10(s+1)}{(s+1)^2+1} \ .$$

Hence

$$(s^2-s-2)\bar{x} = (s-2)(s+1)\bar{x} = s+2+\frac{10(s+1)}{s^2+2s+2} \Rightarrow \bar{x} = \frac{s+2}{(s+1)(s-2)} + \frac{10}{(s-2)(s^2+2s+2)}$$

Using partial fractions,

$$\frac{s+2}{(s+1)(s-2)} = \frac{4}{3}\frac{1}{s-2} - \frac{1}{3}\frac{1}{s+1}$$

and taking the inverse transforms of the last two, and combining with (ii), we have

$$x(t) = \frac{7}{3}e^{2t} - e^{-t}\left(\cos t + 3\sin t + \frac{1}{3}\right) \ .$$

4. (a) A system of linear equations is given as

$$\begin{array}{rcrcrcl} -x & + & y & + & 2z & = & 0\,, \\ x & + & \alpha y & - & z & = & -3\,, \\ 2x & + & y & - & 6z & = & \beta\,, \end{array}$$

where $\alpha, \beta \in \mathbb{R}$ are parameters.

(i) Write the system in matrix form $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$, find the determinant of the matrix A, and give the value of α for which A is singular. [3]

Solution: In matrix form, the system is

$$A\underline{\mathbf{x}} = \begin{pmatrix} -1 & 1 & 2 \\ 1 & \alpha & -1 \\ 2 & 1 & -6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \\ \beta \end{pmatrix} = \underline{\mathbf{b}}\,.$$

The determinant of A, expanding by the second row is

$$|A| = -1(-6-2)+\alpha(6-4)-(-1)(-1-2) = 5+2\alpha$$

and the value for which A is singular is when $\det(A) = 0 \Rightarrow \alpha = -5/2$.

(ii) Let α be the value found in (i). Use Gaussian elimination to reduce the augmented coefficient matrix $(A : \underline{\mathbf{b}})$ to echelon form, and find $N(A)$, the nullspace of A. What kind of geometrical object does the nullspace represent? [5]

Solution: Begin with the augmented coefficient matrix and let the first row operations be $2R_2$, to eliminate the fraction, and $-R_1$ to get pivot $= 1$:

$$(A : \underline{\mathbf{b}}) = \begin{pmatrix} -1 & 1 & 2 & : & 0 \\ 1 & -5/2 & -1 & : & -3 \\ 2 & 1 & -6 & : & \beta \end{pmatrix} \underset{2R_2}{\overset{}{\sim}} \begin{pmatrix} 1 & -1 & -2 & : & 0 \\ 2 & -5 & -2 & : & -6 \\ 2 & 1 & -6 & : & \beta \end{pmatrix} \underset{R_3-2R_1}{\overset{R_2-2R_1}{\sim}} \begin{pmatrix} 1 & -1 & -2 & : & 0 \\ 0 & -3 & 2 & : & -6 \\ 0 & 3 & -2 & : & \beta \end{pmatrix}$$

$$\underset{R_3+R_2}{\sim} \begin{pmatrix} 1 & -1 & -2 & : & 0 \\ 0 & -3 & 2 & : & -6 \\ 0 & 0 & 0 & : & \beta-6 \end{pmatrix}$$

The nullspace of A is all solutions of $A\underline{\mathbf{x}} = \underline{\mathbf{0}}$. Hence the second row has $-3y+2z = 0$. Let the free variablr be $z = \lambda \in \mathbb{R}$, so that $y = \frac{2}{3}z$. The first row now has $x - y - 2z = x - \frac{2}{3}\lambda - 2\lambda = 0 \Rightarrow x = \frac{8}{3}\lambda$. The nullspace, in

$$N(A) = \left\{ \lambda \begin{pmatrix} 8/3 \\ 2/3 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R} \right\}.$$

This is a line through the origin in $\mathbb{R}^3$.

(iii) Find the value of β for which the system of equations is consistent, explaining why this is necessary. Let β be take this value. Now find a particular solution of the system, and hence the general solution, giving a geometric interpretation of the solution. [5]

Solution: We require $\beta = 6$ for the system to be consistent. For any other value, e.g. $\beta = 7$, the third row would give the equation $0x+0y+0z = 1$, for which there are no solutions: the system would be inconsistent.

A particular solution is most easily found setting the free variable equal to zero, so let $z = 0$. The second row now has $-3y + 2z = -6 \Rightarrow y = 2$. Row one has $x - y - 2z = 0 \Rightarrow x = 2$, and the particular solution is

$$\underline{\mathbf{x}}_p = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}$$

so the general solution is the sum of particular solution and nullspace:

$$\underline{\mathbf{x}} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 8/3 \\ 2/3 \\ 1 \end{pmatrix},$$

a line in $\mathbb{R}^3$, not through the origin, and parallel to the line given by the nullspace.

(b) A matrix is given as

$$A = \begin{pmatrix} 1 & -1 & 1 & -1 \\ 2 & -3 & 4 & 2 \\ -1 & 3 & -5 & -7 \end{pmatrix}.$$

Obtain the echelon form of A, and state its rank. Hence find a basis for each of the four fundamental subspaces of A: nullspace, column space, row space, and left nullspace. In each case, give your reasoning! [12]

Solution: Using row operations

$$\begin{pmatrix} 1 & -1 & 1 & -1 \\ 2 & -3 & 4 & 2 \\ -1 & 3 & -5 & -7 \end{pmatrix} \underset{\substack{R_2 - 2R_1 \\ R_3 + R_1}}{\sim} \begin{pmatrix} 1 & -1 & 1 & -1 \\ 0 & -1 & 2 & 4 \\ 0 & 2 & -4 & -8 \end{pmatrix} \underset{R_3 + 2R_2}{\sim} \begin{pmatrix} 1 & -1 & 1 & -1 \\ 0 & -1 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

is the echelon form, and the rank is two. The columns containing pivots are independent. These are the first two, so the column space has a basis given by the first two columnns of A:

$$\text{basis for } C(A) \ : \ \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} -1 \\ -3 \\ 3 \end{pmatrix}.$$

In echelon form we have zero entries below the pivots, so no good to give linear combinations: we use the original columns of A, Gaussian elimination merely identifies which are pivot columns.

Similarly the rows of the echelon form containing pivots are a basis for the row space:

$$\text{basis for } C(A^T) \ : \ \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 0 \\ -1 \\ 2 \\ 4 \end{pmatrix}.$$

We need to use pivot rows of the echelon form, as these are independent. The original rows might not be independent.

To find a basis for the nullspace, we solve $A\underline{\mathbf{x}} = \underline{\mathbf{0}}$. There are two free variables, x_3 and x_4, which we set equal to $0, 1$ and $1, 0$ to find the independent special solutions of $A\underline{\mathbf{x}} = \underline{\mathbf{0}}$ which give the basis. Hence, let $x_3 = 1$ and $x_4 = 0$. Then row two gives $-x_2 + 2x_3 + 4x_4 = 0 \Rightarrow x_2 = 2$ and row one gives $x_1 - x_2 + x_3 - x_4 = 0 \Rightarrow x_1 = 1$ and we have $\underline{\mathbf{x}}_1^s = (1, 2, 1, 0)^T$.

Similarly, let $x_3 = 0$ and $x_4 = 1$. Then row two has $-x_2 + 2x_3 + 4x_4 = 0 \Rightarrow x_2 = 4$ and row one reads as $x_1 - x_2 + x_3 - x_4 = 0 \Rightarrow x_1 = 5$, giving the second special solution $\underline{\mathbf{x}}_2^s = (5,4,0,1)^T$. So we have

$$\text{basis for } N(A) \ : \ \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 5 \\ 4 \\ 0 \\ 1 \end{pmatrix}.$$

Finally, a basis for the left nullspace $N(A^T)$ is obtained by applying the same row operations to the identity as to A:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \underset{\substack{R_2-2R_1 \\ R_3+R_1}}{\sim} \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \underset{R_3+2R_2}{\sim} \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & 2 & 1 \end{pmatrix} = E$$

We reason as follows: $\underline{\mathbf{y}} \in N(A^T)$ means

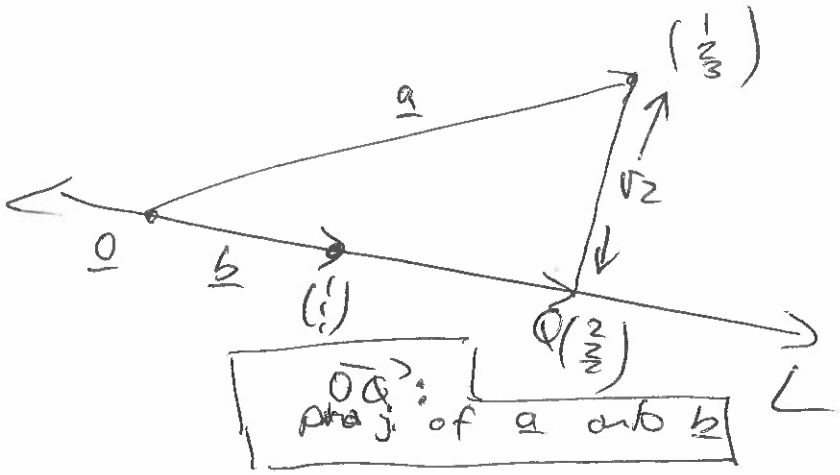
$$A^T\underline{\mathbf{y}} = \underline{\mathbf{0}} \Rightarrow (A^T\underline{\mathbf{y}})^T = \underline{\mathbf{0}}^T \Rightarrow \underline{\mathbf{y}}^T A = \underline{\mathbf{0}}^T$$

so an element of $N(A^T)$ is one which,as a row, multiplying A on the left gives a zero row in the echelon form.

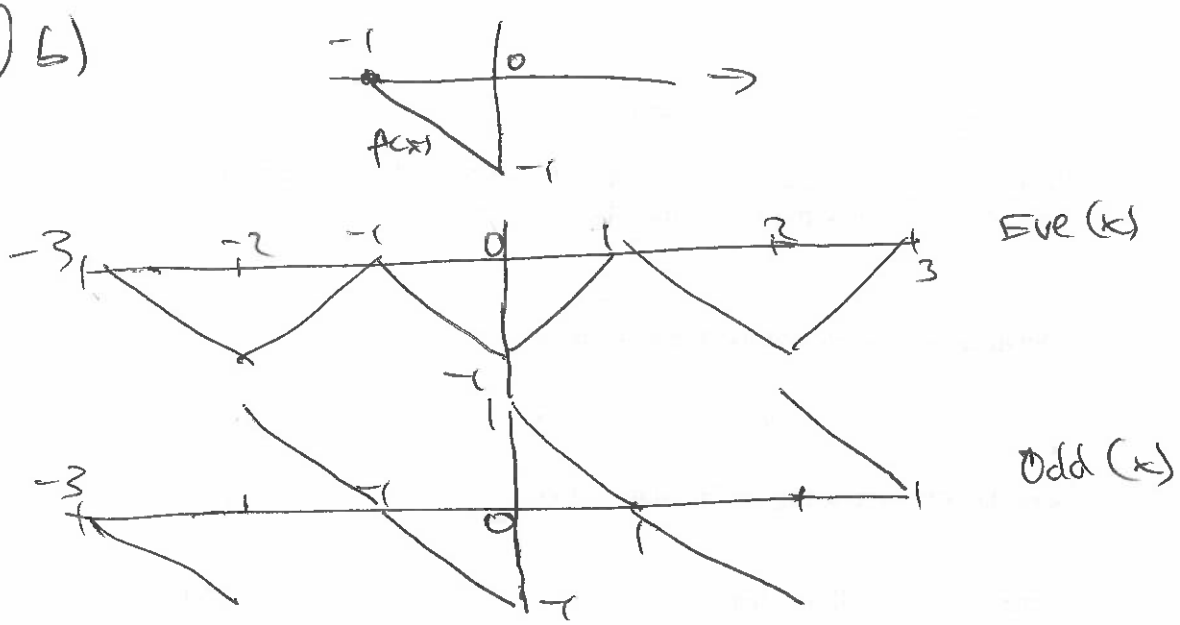
Multiplying A on the right by E gave the echelon form, so the action of the last row of E gave the zero row in the echelon form of A, so this is the basis for the left nullspace:

$$\text{basis for } N(A^T) \ : \ \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix}.$$

(1) a)


$\begin{pmatrix}1\\2\\3\end{pmatrix}$
$\underline{a}$
$\underline{v}_2$
$\underline{0}$
$\underline{b}$
$\begin{pmatrix}1\\1\end{pmatrix}$
Q $\begin{pmatrix}2\\2\end{pmatrix}$
$\overrightarrow{OQ}$: proj of $\underline{a}$ onto $\underline{b}$
L

(2) b)


-1
0
f(x)
-1
-3
-2
-1
0
1
2
3
Eve (x)
-1
1
-3
1
0
-1
Odd (x)